HACK FOR SOCIAL CAUSE · EDTECH

Live prototype · Built for Bharat

Student Setu

Bridging Students, Building Futures



studentsetuconnect.netlify.app

COVER PAGE

Hack for Social Cause

PROBLEM STATEMENT / CONCEPT TITLE

Student Setu — AI-powered Curriculum-to-Career Bridge

Personalised skilling for students, an AI copilot for teachers and vocational training in Hindi — built to work on low-bandwidth phones across Bharat.

THEME

Quality Education & Skill Development

Remote learning · personalised education · empowering teachers · vocational training

TEAM NAME (REGISTERED ON PORTAL)

‹Team Name›

TEAM ID (REGISTERED ON PORTAL)

‹Team ID›

STATE / UT

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Scan for live demo

‹Team Name›

Problem – Proposed Solution

Hack for Social Cause

THE CORE PROBLEM — EDUCATION ISN'T TURNING INTO JOBS

45% of Indian graduates are not job-ready

India Skills Report 2025 (54.81% employable)

~4% of Indians aged 15–59 have formal vocational training

PLFS 2023-24, MoSPI

27:1 students per teacher — little time for prep or personal attention

AISHE 2021-22, Ministry of Education

26% of Web Developer job skills covered by a B.Tech syllabus, even at graduation

Our gap-map analysis (RGPV B.Tech)

PROPOSED SOLUTION — ONE PLATFORM, THREE PILLARS

Setu Skill Bridge

For students

- Gap Map: real syllabus (Course → University → Semester) vs 13 job roles × 45 skills
- Personalised week-by-week plan from free courses + AI coach tips
- Verified Skill Badges: server-graded, signed, checkable by any employer

Solves: skills gap · personalised education

Teacher Copilot

For teachers

- AI lesson plans & question papers from the syllabus, in English or Hindi
- Auto-graded class quizzes — students join with a code, no login
- Weak-topic heatmap → one-click re-teach; verified teacher accounts

Solves: empowering teachers · quality education

Bharat-first Access

For every learner

- 7 vocational trade tracks in Hindi, linked to Skill India, Apprenticeship & ITI
- Lite mode for 2G/3G, offline installable app, listen-aloud audio
- WhatsApp sharing, full Hindi UI; hostel & tiffin finder as optional support

Solves: remote learning · vocational training

INNOVATION & UNIQUENESS

- **Syllabus-mapped gap map, semester by semester**
- **Tamper-proof badges: answers never reach the browser**
- **Closed teaching loop: quiz → heatmap → re-teach**
- **Works without AI & offline: templates + PWA**

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Technical Approach

Hack for Social Cause

TECHNOLOGIES — SERVERLESS, LOW-COST STACK

Frontend (PWA) — React 18 · Vite 6 · Tailwind CSS · Framer Motion · React Router · Service Worker · Web Speech API

Serverless API — Netlify Functions (Node 20, TypeScript): copilot · badge · classroom · chat

AI — Google Gemini 2.5 Flash (JSON mode)

Data — Netlify Blobs (class quizzes & results)

Security — HMAC-SHA256 badges, keys on server

METHODOLOGY — HOW IT WORKS

Student journey

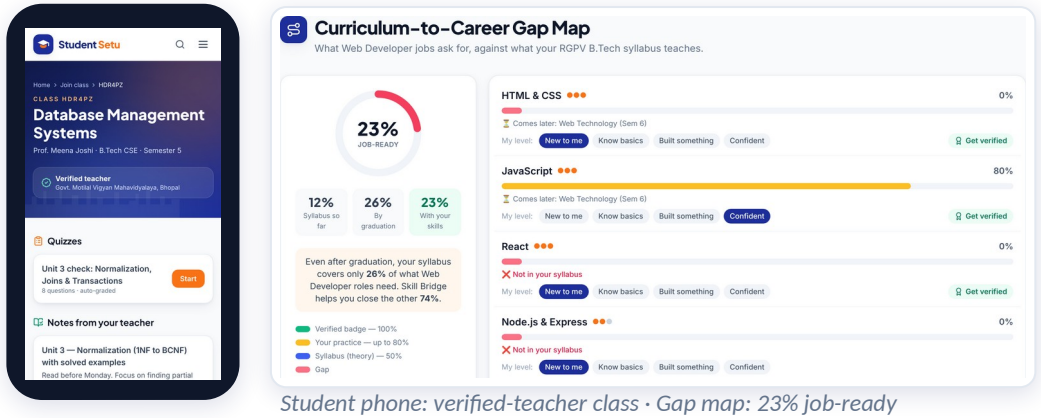
Pick course · univ · sem → Gap map → Weekly plan → Server-graded test → Signed badge

Teacher loop

Syllabus topic → AI lesson, paper or quiz → Share code on WhatsApp → Students submit, no login → Heatmap → re-teach

Process: agile build–test sprints · Playwright end-to-end tests · GitHub → Netlify auto-deploy

WORKING PROTOTYPE — REAL SCREENS



Student phone: verified-teacher class · Gap map: 23% job-ready

GitHub
prakharporwal518-collab/Student-Setu

Live demo
studentsetuconnect.netlify.app

Demo video
‹Add demo video link›

Extent of AI usage

In product: Gemini writes lesson plans, papers, quizzes, coach tips & chat replies
Guardrails: strict JSON schema, key on server, “review before use”, template fallback
No AI where trust matters: quiz grading & badge signing are deterministic
In development: Claude Code AI assistant for coding & testing, reviewed by the team

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Impact Potential

POTENTIAL IMPACT ON THE TARGET AUDIENCE

Students

4.33 Cr in higher education

Know the exact job-skill gap from Semester 1 · free personalised plan · skill proof employers can verify

Teachers

15.98 L faculty

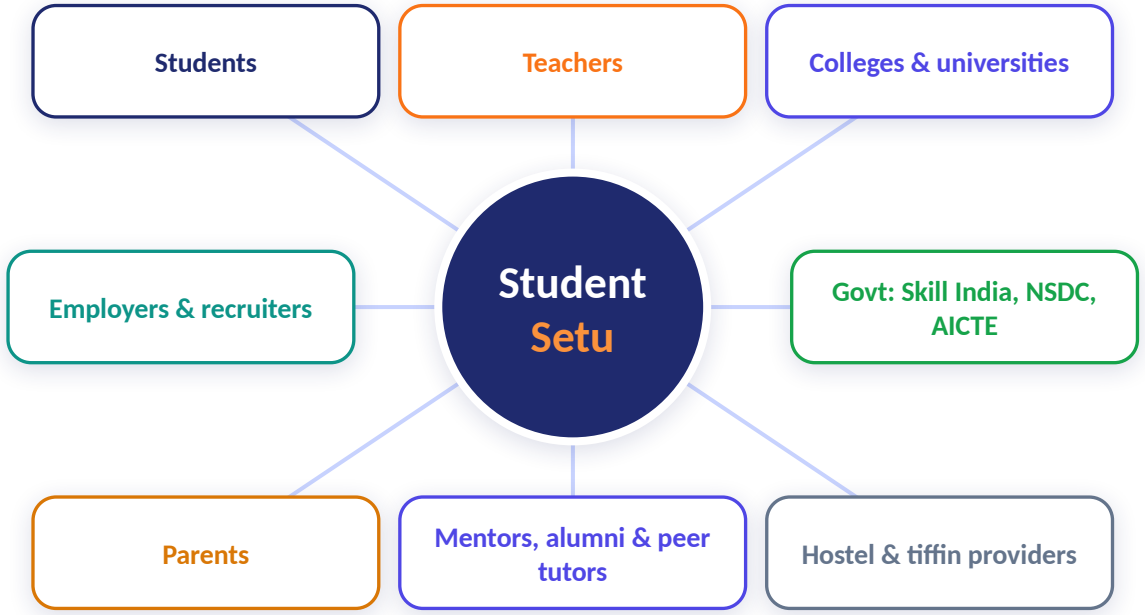
Lesson plans, papers & quizzes in minutes · see what the class didn't understand · re-teach with data

Vocational learners

~96% lack formal training

Trade skills in Hindi on a basic phone · tests & badges · links to Skill India, Apprenticeship India & ITI

STAKEHOLDERS



BENEFITS OF THE SOLUTION

Social

Equal access for rural, Hindi-medium & first-generation learners; safe stay & food info for migrant students

Economic

₹0 for learners, free courses, verified skills → better jobs & wages; saves teachers' prep hours

Educational

Topic-level outcome data; aligned with NEP 2020 — multilingual, vocational & competency-based

Institutional

Colleges see syllabus-vs-industry gaps to update curriculum; employers hire on verified skills

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Feasibility and Risk Analysis

FEASIBILITY — ALREADY PROVEN BY A LIVE PROTOTYPE

Technical

Live app + 4 serverless functions; every flow tested end-to-end

Economic

~₹0 at pilot scale on free tiers (Netlify, Gemini); no app-store cost

Operational

Runs in any phone browser; students need no login; English + Hindi

Scalable

Serverless auto-scaling; a new university = one new syllabus file

POTENTIAL CHALLENGES & RISKS → STRATEGIES TO OVERCOME THEM

Risk / challenge	Level	Mitigation strategy
AI gives wrong or low-quality content	High	"Review before use" label · teacher edits all output · strict JSON schema · template fallback
Poor internet & low-end phones	High	Lite mode (no images or animations) · offline installable PWA · lazy-loaded pages · audio
Fake teacher accounts	Medium	Admin verifies college + employee ID · only verified teachers can run classes
Badge cheating or forgery	Medium	Answers never sent to the browser · server grading · HMAC-SHA256 signature · public verify link
Low teacher adoption	Medium	3 free tries without login · Hindi · WhatsApp sharing · saves prep time from day one
Keeping syllabus data current at scale	Medium	One structured file per university · next: university & teacher uploads, college partnerships
Student data privacy	Low	Students join with class code + name only · no phone or email · API keys only on the server

‹Team Name›

Team

TEAM COMPOSITION & ROLES

‹Member 1›
Team Lead · Full-stack

Architecture, serverless functions, deployment

‹Member 2›
Frontend & UI/UX

React screens, Hindi UI, Lite mode, accessibility

‹Member 3›
AI & Backend

Gemini prompts, quiz grading, signed badges

‹Member 4›
Research & Content

Syllabus–skill mapping, vocational modules, testing

CHALLENGES WE SOLVED

- **Keeping test answers secret in a web app**
  → grading moved to the server + signed badges
- **AI must never break a live class**
  → strict JSON output + ready-made templates
- **Learners on 2G/3G connections**
  → Lite mode + offline installable app
- **Mapping syllabus to real job skills**
  → 45-skill × 13-role dataset, semester-wise
- **Trusting who is really a teacher**
  → admin verification with college & employee ID

OUR JOURNEY

1 **Student-life MVP**

Hostels, tiffin, colleges, local connect

2 **Re-read the problem**

Pivoted to learning-first; life support kept optional

3 **Setu Skill Bridge**

Gap map, personal plan, verified badges

4 **Teacher Copilot**

Lessons, papers, quizzes, class heatmap

5 **Bharat add-ons**

Hindi, Lite mode, offline app, vocational

6 **Trust & launch**

Teacher verification, live on Netlify

‹Team Name›

References

Data & research

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Government platforms we link to

Skill India Digital Hub
skillindiadigital.gov.in

Apprenticeship India
apprenticeshipindia.gov.in

SWAYAM & NPTEL free courses
swayam.gov.in · nptel.ac.in

National Scholarship Portal
scholarships.gov.in

Technology documentation

Google Gemini API
ai.google.dev

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react.dev · vite.dev · tailwindcss.com

Web Speech API & PWAs (MDN)
developer.mozilla.org

Our project

- **Live demo:** studentsetuconnect.netlify.app
- **Source code:** github.com/prakharporwal518-collab/Student-Setu
- **Demo video:** ‹Add demo video link›

Live demo

GitHub